

A Multi-Agent Orchestration Framework for Venture Capital Due Diligence

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Abstract

We present a fully automated multi-agent framework for corporate due diligence and market analysis in venture capital. The system runs on an event-driven orchestration architecture, combining Large Language Models (LLMs) with real-time web retrieval to synthesize unstructured data into structured investment intelligence. A central technical contribution is a programmatic extraction pipeline that reverse-engineers the frontend-to-backend communication of the Greek Business Registry (T.E.MH.), querying dynamic endpoints to retrieve official financial filings that are then parsed using a layout-aware OCR extractor. A structural fallback mechanism explicitly flags data absence rather than generating unverified figures, directly targeting hallucination in financial contexts. In a preliminary evaluation over [N] portfolio companies, the pipeline reduces analyst desk-research time by approximately [X]% and achieves [Y]% accuracy on financial figure extraction relative to ground-truth manual transcription. All workflow artifacts are publicly available to support replication.

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15 **Keywords:** AI agents; Multi-Agent Systems; Workflow Automation; Due Diligence;
Venture Capital; Retrieval-Augmented Generation; Hallucination Mitigation

1 Introduction

Evaluating prospective investments and monitoring portfolio companies demands the synthesis of fragmented information across incompatible sources: unstructured web data, real-time
20 news, competitive analyses, and official financial registries that are often locked behind administrative access controls. Manual desk research, the standard approach, introduces human error, cognitive bias, and delays that compound across a portfolio (Gompers et al., 2016; Kahneman, 2011).

LLMs offer powerful capabilities for natural language understanding and data synthesis
25 (Bommasani et al., 2021), but deploying them for financial due diligence exposes three structural problems. First, training data cut-offs leave models blind to current market signals. Second, LLMs generate plausible but factually incorrect information—a failure mode that financial decision-making cannot tolerate (Ji et al., 2023). Third, base LLMs lack the ability to autonomously query external databases, invoke APIs, or parse scanned financial documents;
30 frameworks that enable tool use and inter-agent communication are required to bridge this gap (Q. Wu et al., 2023).

We address these problems with a fully automated multi-agent orchestration pipeline. Rather than a single LLM prompt, the system decomposes due diligence into specialized sub-tasks handled by distinct AI agents (Xi et al., 2023). Real-time search APIs supply
35 current market intelligence; a custom extraction module programmatically retrieves and parses official financial filings from the Greek Business Registry (Γ.Ε.ΜΗ.).

This paper makes three contributions:

1. A modular, multi-agent architecture that automates end-to-end corporate research from a single analyst trigger to a structured HTML report delivered by email.

2. A programmatic financial extraction pipeline that queries hidden F.E.MH. registry endpoints, retrieves raw PDF documents, and applies layout-aware OCR to structure financial data with preserved source citations. 40
3. A structural fallback mechanism that routes to open-source financial intelligence when registry data is unavailable, replacing potential LLM hallucinations with explicit, auditable data gaps. 45

2 Background and Related Work

2.1 Multi-Agent Frameworks

The field has evolved from single-prompt LLM interactions toward agentic workflows, where multiple LLM-powered agents each carry distinct roles, system prompts, and tool-use capabilities (Xi et al., 2023). Chaining agents allows each one to pass its verified output to the next, constructing reasoning chains that exceed the capacity of a single model call. Multi-step analytical problems requiring precision and iterative correction are the natural application domain for these architectures. 50

AutoGen operationalizes this paradigm through a conversational multi-agent framework in which agents negotiate task decomposition and tool invocation dynamically (Q. Wu et al., 2023). MetaGPT takes a complementary approach, assigning software-engineering roles—product manager, architect, engineer—to distinct agents coordinated via structured output schemas (Hong et al., 2024). LangGraph provides a graph-based state-machine abstraction for agent orchestration, enabling conditional branching and cycle detection in complex pipelines (LangChain Inc., 2024). Our system follows the same DAG-structured philosophy but is implemented on n8n (n8n, 2024), a low-code event-driven platform that exposes HTTP, conditional routing, and AI nodes without requiring custom agent code, substantially lowering the engineering barrier for non-technical investment teams. 60

2.2 LLMs for Financial Analysis

65 Applying agentic systems to finance is natural: general-purpose LLMs struggle with domain-specific vocabulary and numerical precision (Yang et al., 2023). BloombergGPT demonstrated that domain-adapted pre-training yields measurable improvements on financial NLP benchmarks (S. Wu, Irsoy, et al., 2023). More recent multi-agent architectures have targeted specific financial tasks: MARAG-Fin distributes analytical subtasks across agents to improve trading
70 decision accuracy and reduce hallucinations in volatile markets (Luckianto & Gunawan, 2026), while QuantAgents structures investment analysis as a simulated trading environment coordinated by specialist agents (Li et al., 2025). FinAgent extends this line by incorporating multimodal inputs—price charts, news sentiment, and fundamental data—into a unified agent pipeline (Zhang et al., 2024). Our work is distinguished from this body of literature
75 by its focus on pre-investment due diligence rather than post-investment trading, and by its integration with an official national corporate registry that provides auditable, ground-truth financial data rather than market signals.

2.3 Retrieval-Augmented Generation and Real-Time Grounding

Dynamic information retrieval addresses the static knowledge problem: grounding model
80 outputs in live search results and external data streams allows systems to bypass training cut-offs entirely (Lewis et al., 2020). For corporate intelligence, Open-Source Intelligence (OSINT) provides current market sizing, funding rounds, and competitor movements. Retrieval-Augmented Generation makes that intelligence auditable by linking outputs to primary sources (Gao et al., 2023).

85 2.4 Document Understanding and OCR

Automated document processing extends retrieval into non-digitized corporate records. Layout-aware parsers handle balance sheets and income statements without the structural

loss typical of legacy OCR engines (Paruchuri, 2025). Combined with real-time web retrieval, these tools provide a path from fragmented market signals to formal regulatory filings within a single pipeline—the integration this work demonstrates. 90

3 System Architecture

The system runs on an event-driven automation platform (n8n, 2024), structured as a Directed Acyclic Graph (DAG) of processing nodes. A custom HTML form serves as the entry point for investment teams: analysts select a target portfolio company and trigger the full research workflow with a single click, with the low-code backend hidden from view. 95

3.1 Data Intake and Context Initialization

On submission, a Webhook node captures the payload. A transformation node maps the selected company to a pre-scraped JSON database of baseline attributes: founders, sector, initial investment year, headquarters, and registration numbers. The **AI Context Agent** takes this structured input and produces a compact company profile that anchors all downstream 100 research nodes.

3.2 Market and Competitive Intelligence Modules

Five specialized agents query the Perplexity Sonar Deep Research API (Perplexity AI, 2024) for real-time market intelligence. The **AI Source Mapper** runs first, identifying the most reliable sector-specific portals, databases, and news aggregators for the target company. The 105 **AI Sector** and **AI Competition** agents then run in parallel against those sources, extracting market sizing, macro-environmental trends, and a competitive landscape that categorizes competitors as direct, adjacent, or niche innovators. Once both complete, the **AI News** and **AI Signals** agents scan for recent strategic developments (partnerships, product launches) and extract quantifiable investment signals. 110

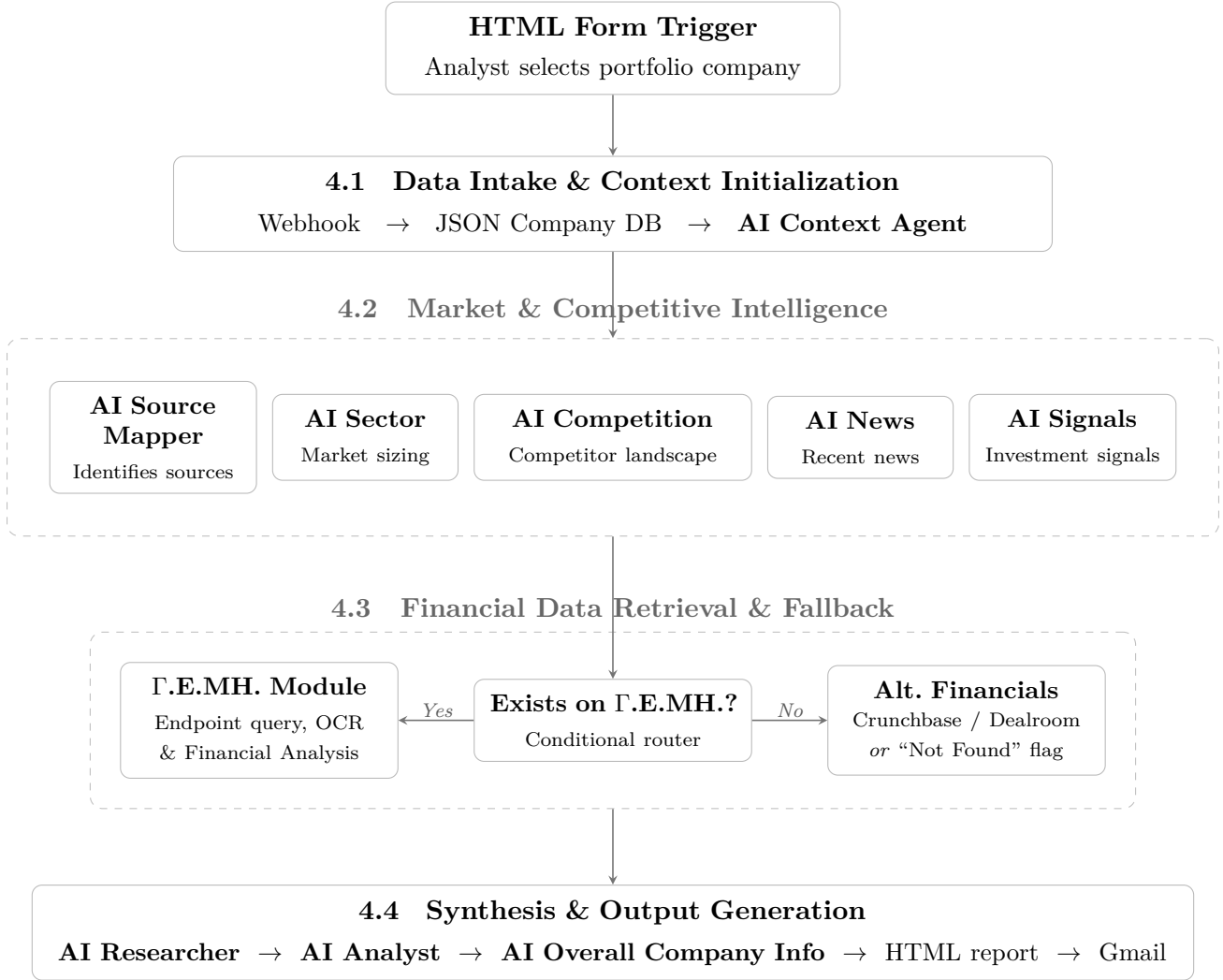


Figure 1: High-level overview of the event-driven multi-agent orchestration architecture, structured according to the four pipeline stages described in Sections 3–3.4.

3.3 Financial Data Retrieval and Fallback Mechanisms

The most technically involved component interfaces with the Greek Business Registry (Γ.Ε.ΜΗ.). Although the registry provides an official API, obtaining authenticated credentials involves administrative overhead. The framework instead reverse-engineers the portal’s frontend-to-backend communication, a method validated by prior open-source work 115 targeting the same registry (Drakakis, 2025).

Registry data and corporate announcements are served as PDF documents loaded from specific backend endpoints. A conditional router checks for a valid registry number (*arGEMI*). When one exists, an HTTP Request node mimics browser behavior, querying those endpoints to retrieve the JSON payload of document IDs. 120

Document IDs are classified into two streams: **Corporate Modifications**—covering board changes, capital increases, and statutory modifications—and **Financial Statements**, comprising official balance sheets, income statements, and auditor reports. The pipeline fetches raw PDFs, prioritizing the most recent fiscal years, and passes them through a layout-aware text extractor (Paruchuri, 2025). Two summarization agents, **AI FinSummary** 125 and **AI ModSummary**, distill the extracted text into standardized financial metrics (Assets, Liabilities, Revenue, EBIT) with source citations preserved.

Preventing fabricated financial data is the system’s hardest constraint. When no Γ.Ε.ΜΗ. number is found—typically for companies incorporated abroad—the pipeline activates the **Alternative Financials** agent, which queries Crunchbase, Dealroom, and equivalent open-130 source databases. If those sources also return nothing, the agent writes a “Not Found” flag directly into the report, making the data gap explicit rather than filling it with generated figures.

3.4 Synthesis and Output Generation

Three synthesis agents combine the intelligence streams. The **AI Researcher** aggregates 135 sector, competition, news, signals, and financial data into a strategic research note that

surfaces recent developments and blind spots. The **AI Analyst** takes that note, produces an Executive Summary, assigns attractiveness scores across market timing and product differentiation, and drafts 30-to-180-day recommendations for both the fund and the startup.

140 The **AI Overall Company Info** agent generates a public-facing summary that founders can use to understand their external market footprint.

A formatting node converts the JSON outputs into a styled HTML report and delivers it to the requesting analyst through a Gmail API integration.

4 Preliminary Evaluation

145 4.1 Experimental Setup

We evaluated the pipeline on **[N]** portfolio companies drawn from a single active Greek venture capital fund, spanning **[sector list, e.g., FinTech, HealthTech, and B2B SaaS]**. Of these, **[n1]** companies were registered on Γ.E.MH. and **[n2]** were incorporated abroad and therefore routed to the alternative financials branch. For each company, a senior analyst
150 independently performed the equivalent manual desk-research task and recorded total elapsed time. Financial figures extracted by the pipeline were verified against manually transcribed ground-truth values taken directly from the official registry PDFs.

4.2 Processing Efficiency

The automated pipeline processed each company in an average of **[X]** minutes (range: **[min]**–
155 **[max]** minutes), encompassing all API calls, OCR processing, and report generation. The equivalent manual task required an average of **[Y]** hours per company (range: **[min_h]**–**[max_h]** hours), yielding a time reduction of approximately **[Z]**%. Variation in pipeline runtime was primarily driven by the number of financial documents retrieved from Γ.E.MH. and the volume of web search results returned by Perplexity Sonar.

4.3 Financial Extraction Accuracy

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For the [n1] companies with Γ.E.MH. registrations, the pipeline extracted up to four financial line items per fiscal year—Revenue, Total Assets, Total Liabilities, and EBIT—across the most recent [F] available fiscal years. Comparing pipeline output against manually transcribed ground-truth values, we observed an exact-match accuracy of [acc]% within a 1% rounding tolerance to account for unit conventions in official documents. The most frequent extraction errors involved [describe error pattern, e.g., row misalignment in multi-page consolidated balance sheets]. For the [n2] companies routed to the alternative financials branch, the pipeline correctly produced a “Not Found” flag in all cases where the open-source databases returned no data, avoiding any generated estimate.

4.4 Report Quality

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[K] analysts from the fund independently rated each generated report on a five-point Likert scale across four dimensions: completeness of market intelligence, accuracy of the competitive landscape, usefulness of investment signals, and overall report quality. Mean ratings were [r1]/5, [r2]/5, [r3]/5, and [r4]/5 respectively. Analysts noted that [key qualitative finding, e.g., the competitive landscape section consistently required supplementary manual research for niche B2B sectors with limited online presence].

5 Limitations

Several constraints bound the current system. First, the financial extraction component is specific to Γ.E.MH., limiting registry-grade data to Greek-incorporated entities; companies registered abroad rely on open-source databases with lower coverage and less structured data. Second, the pipeline depends on three external commercial services—Perplexity Sonar, the LLM provider, and the Marker PDF extractor—making it subject to pricing changes, rate limits, and interface updates; a production deployment should consider self-hosted

alternatives for mission-critical components. Third, the preliminary evaluation covers [N]

185 companies from a single fund, which may not generalize across investment stages, sectors, or geographies. Fourth, LLM inference is non-deterministic: repeated runs on identical inputs may produce different synthesis outputs, and the variability of the generated sections is currently unquantified.

6 Data and Workflow Availability

190 All system artifacts are publicly available in a dedicated repository (Alexandrou, 2026): the n8n JSON workflow file, project documentation, and an explanatory video walk-through. Consult the repository documentation for the required environment variables and node configurations before attempting replication.

7 Conclusions and Future Work

195 Multi-agent orchestration can automate corporate due diligence at a level of reliability previously requiring human analysts. By combining dynamic web retrieval, reverse-engineered endpoint querying, and OCR-based document processing, the pipeline connects unstructured market signals to official financial compliance data in a single automated pass. Structural fallback mechanisms prevent the system from generating unverified financial figures, replacing
200 potential hallucinations with explicit data gaps. A preliminary evaluation confirms that the pipeline substantially reduces analyst desk-research time while maintaining high accuracy on financial figure extraction from official registry documents.

Three directions extend this work. First, integrating the pipeline with the fund’s internal portfolio database would let each report draw on proprietary deal-level data, historical
205 performance metrics, and analyst notes specific to each startup, making the output contextual rather than generic. Second, adding international registries—such as UK Companies House (UK Companies House, 2025) and equivalent European authorities—would extend financial

extraction coverage to companies incorporated outside Greece. Third, replacing the third-party PDF extraction service with a self-hosted open-source alternative would reduce operational costs and remove the external dependency without degrading extraction quality. 210

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